

HAYDEN A. JOHNSON

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EDUCATION

- Ph.D., Oceanography** Jun. 2025
Scripps Institution of Oceanography
University of California San Diego *La Jolla, CA, USA*
Ph.D. Thesis: “The release of bubbles and other small-scale processes at the ice-water boundary of marine-terminating glaciers”
Advisor: Grant Deane
Committee: Mike Buckingham, Jamin Greenbaum, David Saintillan, and Fiamma Straneo
- B.S., Physics** (Honours with High Distinction) 2019
University of Toronto *Toronto, ON, Canada*
Honours Thesis: “Integration and calibration of SPIDER pinhole sun sensors”
Advisor: Barth Netterfield

EMPLOYMENT

- Postdoctoral Researcher**
College of Fisheries and Ocean Sciences Oct. 2025 - present
University of Alaska Fairbanks *Fairbanks, AK, USA*
- Graduate Student Researcher**
Scripps Institution of Oceanography 2019-2025
University of California San Diego *La Jolla, CA, USA*

PUBLICATIONS

Peer-reviewed

7. **Johnson, H. A.**, Deane, G. B., Stokes, M. D., Glowacki, O., Moskalik, M., Chitre, M., Vishnu, H., & Powell, C. Potential energy of air bubbles in glacier ice and its conversion into acoustic energy. Manuscript accepted April 2026 for publication in *Journal of the Acoustical Society of America*.
6. Weidner, E., Deane, G. B., Le Boyer, A., Alford, M. H., Vishnu, H., Chitre, M., Stokes, M. D., Glowacki, O., **Johnson, H. A.**, & Straneo, F. (2025). High frequency broadband acoustic systems as a tool for high latitude glacial fjord research. *The Cryosphere*, 19, 4715-4740. <https://doi.org/10.5194/tc-19-4715-2025>.
5. Vishnu, H., Chitre, M., Glowacki, O., Stokes, D., **Johnson, H. A.**, Moskalik, M., & Deane, G. B. (2025). Acoustic activity indicates submarine melt at tidewater glaciers. *Journal of Glaciology*, 71, e83. <https://doi.org/10.1017/jog.2025.10061>.
4. Grossman S., **Johnson, H. A.**, Stokes, M. D., & Deane, G. B. (2024). Toward monitoring submarine glacier melt using hydroacoustics: The role of timescale in the signal of bubble release. *Journal of the Acoustic Society of America*, 156, (3), 1820–1838. <https://doi.org/10.1121/10.0028628>.
3. **Johnson, H. A.**, Glowacki, O., Deane, G. B., & Stokes, M. D. (2024). Brief communication: A technique for making in-situ measurements at the ice-water boundary of small pieces of floating glacier ice. *The Cryosphere*, 18, 265-272. <https://doi.org/10.5194/tc-18-265-2024>.

2. Vishnu, H., Deane, G. B., Glowacki, O., Chitre, M., **Johnson, H. A.**, Moskalik, M., & Stokes, M. D. (2023). Depth-dependence of the underwater noise emission from melting glacier ice. *JASA Express Letters*, 3 (2). <https://doi.org/10.1121/10.0017348>.
1. Godin, P. J., **Johnson, H. A.**, Piunno, R., Le Bris, K., & Strong, K. (2019). Conformational analysis and global warming potentials of 1, 1, 1, 2, 3, 3-hexafluoropropane and 1, 1, 2, 2, 3-pentafluoropropane from absorption spectroscopy. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 225, 337-350. <https://doi.org/10.1016/j.jqsrt.2019.01.003>.

Conference Proceedings

2. Xiaohui, Z., Vishnu, H., Chitre, M., Kalyan, B., **Johnson, H. A.**, & Glowacki, O. (2025). Estimating Iceberg Melt Rates from Video Using Novel View Synthesis. *OCEANS 2025 Brest*, 1-8. <https://doi.org/10.1109/OCEANS58557.2025.11104630>.
1. Jenkins, W., **Johnson, H. A.**, Vakhutinsky, S., Helmberger, M. N., Storheim, E., Sagen, H., & Sandven, S. (2022). Analysis of underwater acoustic data collected under sea ice during the Useful Arctic Knowledge 2021 cruise. *Proceedings of Meetings on Acoustics*, 47 (1). <https://doi.org/10.1121/2.0001574>.

PRESENTATIONS

Conference Talks

6. **Johnson, H. A.**, Deane, G. B., Glowacki, O., Stokes, M. D., Chitre, M., Vishnu, H., Weidner, E., & Gollamudi, K. *Acoustic observations of individual bubble release events from melting glacier ice in an arctic fjord*. AGU Fall Meeting, Washington, DC, USA (Dec. 2024).
5. **Johnson, H. A.**, Deane, G. B., Glowacki, O., Stokes, M. D., Chitre, M., Vishnu, H., & Weidner, E. *Acoustic observations of individual bubble release events from melting glacier ice in an arctic fjord*. Acoustical Society of America Spring Meeting, Ottawa, ON, Canada (May 2024).
4. **Johnson, H. A.**, Deane, G. B., Glowacki, O., & Stokes, M. D. *Direct measurements of submarine melting of floating growlers, and implications for parameterizations of submarine melting*. AGU Fall Meeting, San Francisco, CA, USA (Dec. 2023).
3. **Johnson, H. A.**, Deane, G. B., Stokes, M. D., Glowacki, O., Moskalik, M., Chitre, M., Vishnu, H., & Powell, C. *Understanding the underwater noise of melting glacier ice*. International Glaciological Society Symposium on Maritime Glaciers, Juneau, AK, USA (Jun. 2022).
2. **Johnson, H. A.**, Deane, G. B., Stokes, M. D., Glowacki, O., Moskalik, M., Chitre, M., & Vishnu, H. *Predicting the acoustic energy radiated by melting glacier ice*. Acoustical Society of America Fall Meeting, Seattle, WA, USA (Dec. 2021).
1. **Johnson, H. A.**, Vishnu, H., Deane, G. B., Tuckman, P. J., Glowacki, O., & Moskalik, M. *Depth dependence of acoustic signals produced by bubble release events in melting glacier ice*. Acoustical Society of America Fall Meeting, Virtual (Dec. 2020).

Conference Posters

4. **Johnson, H. A.**, Deane, G. B., Stokes, M. D., Glowacki, O., Moskalik, M., Chitre, M., & Vishnu, H. *Modification of the size distribution of bubbles upon their release from glacier ice into water*. Ocean Sciences Meeting, Glasgow, Scotland, UK (Feb. 2026).
3. **Johnson, H. A.**, Dykstra, S., Crews, L., O'Keefe, N., & Mahoney, A. *Warm water melts grounded keels of landfast sea ice near Utqiagvik, Alaska*. Alaska Marine Science Symposium, Anchorage, AK, USA (Jan. 2026).

2. **Johnson, H. A.**, Deane, G. B., Glowacki, O., Stokes, M. D., Weidner, E., Chitre, M., & Vishnu, H. *Direct measurements of Submarine melting of floating growlers*. Ocean Sciences Meeting, New Orleans, LA, USA (Feb. 2024).
1. **Johnson, H. A.**, Deane, G. B., Glowacki, O., & Stokes, M. D. *Measurements of centimeter-scale temperature gradients at the ice-water boundary of floating glacier ice*. AGU Fall Meeting, Chicago, IL, USA (Dec. 2022).

Invited Talks

2. **Johnson, H. A.**, Dykstra, S., Crews, L., O'Keefe, N., & Mahoney, A. *Warm water melts grounded keels of landfast sea ice near Utqiagvik, Alaska*. University of Alaska Fairbanks, Fairbanks, AK, USA (Feb. 2026).
1. **Johnson, H. A.**, Deane, G. B., Stokes, M. D., Glowacki, O., Chitre, M., Vishnu, H., & Weidner, E. *Testing parameterizations of submarine glacier melt with direct observations*. Oregon State University, Corvallis, OR, USA (Oct. 2023).

FIELDWORK

Under-ice mooring deployment, Utqiagvik, Alaska	Apr. 2026
Fall freeze-up campaign, Utqiagvik, Alaska	Oct.-Nov., 2025
Investigating acoustics of ice melt, Polish Polar Station, Svalbard	Jun.-Jul. 2023
Investigating acoustics of ice melt, Polish Polar Station, Svalbard	Jun.-Jul. 2022
Investigating acoustics of ice melt, Polish Polar Station, Svalbard	Sept. 2021
Useful Arctic Knowledge cruise, <i>KV Svalbard</i>	Jun. 2021
UC Ship Funds cruise, <i>RV Sally Ride</i>	Aug. 2020
SWOT calibration & validation cruise, <i>RV Robert Gordon Sproul</i>	Sept. 2019

TEACHING

Guest Lecturer

Coastal Estuaries March 3 & 5, 2026
 College of Fisheries and Ocean Sciences, UAF
 Instructor: Steven Dykstra

Assistant Instructor

Office of Naval Research Acoustics Bootcamp West November 4-8, 2024
 Scripps Institution of Oceanography, UCSD
 Instructors: Grant Deane & Dale Stokes

Teaching Assistant

SIO 203A/MAE 294A: Introduction to Applied Mathematics (graduate level) Fall 2022
 Scripps Institution of Oceanography & Mechanical and Aerospace Engineering Dept., UCSD
 Instructor: Stefan Llewellyn Smith

SIO 175: Analysis of Atmospheric and Oceanographic Data (undergraduate level) Fall 2020
 Scripps Institution of Oceanography, UCSD
 Instructor: Mark Merrifield

MAT 135: Calculus I(A) (undergraduate level) Fall 2017 & Fall 2018
MAT 136: Calculus I(B) (undergraduate level) Spring 2018 & Spring 2019
 Department of Mathematics, University of Toronto
 Instructor: Sarah Mayes-Tang

MENTORING & LEADERSHIP

Research Mentor

Kavi Gollamudi, Marine Physical Laboratory Summer Intern Summer 2024

Peer Mentoring

SIO Peer Mentor Program Leadership Team Member 2021-2022

Peer mentor to Aidan Blaser 2021-present

SIO Graduate Student Council

Physical Oceanography Curricular Group Representative 2021-2022

Seminar Organizing Team Member

Climate, Atmospheric Science, and Physical Oceanography Seminar 2020-2021

HONOURS & AWARDS

Outstanding Student Presentation Award

Fall Meeting 2024

American Geophysical Union

Undergraduate Student Research Award

2018

Natural Sciences and Engineering Research Council of Canada

QUALIFICATIONS & SKILLS

Small Boat Operation

Scripps Institution of Oceanography Motorboat Operator Training Course 2023

California Boater Card 2022

Ontario Pleasure Craft Operator Card 2008

All-Terrain Vehicle Operation

UIC Science ATV Orientation 2025

VHF Radio

Canadian Power and Sail Squadron Restricted Operator Certificate (Maritime) 2015

Firearm Safety

Poway Gun Range Basic Rifle Instruction 2023

California Firearm Safety Certificate 2021

Computer Programming

Data Analysis MATLAB & Python

Microcontroller Software Development C++

Typesetting L^AT_EX